

NEO-TEEM Black Holes & Lensing

Information-Density Structure of Horizons, Shadows, and Spatial Refraction

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Abstract

Black holes in the NEO-TEEM framework are not geometric singularities but finite-information-density structures formed by the collapse of spatial information capacity R . The horizon emerges as a boundary layer where the ratio R/m undergoes its steepest spatial transition, suppressing vibrational evolution and producing the effective freezing of time. The interior is a saturated informational phase in which R and m reach finite limiting values, eliminating singularities and resolving the information-loss problem. Gravitational lensing arises from spatial refraction in a gradient-index medium defined by $n(r) = R_{\text{vac}}/R(r)$, unifying optical refraction and gravitational bending under a single informational principle. Hawking-like radiation originates from slow reconfiguration of the boundary-layer information structure rather than pair creation, yielding a nearly thermal spectrum that preserves information. The theory reproduces all observational successes of General Relativity while predicting subtle deviations in shadow radius, photon-ring asymmetry, frequency-dependent lensing, Shapiro delay, and gravitational-wave echoes. Black holes and cosmic expansion are shown to be opposite manifestations of the same informational dynamics: local collapse of R versus global growth of R . These results establish NEO-TEEM as a unified, physically transparent, and observationally testable framework for black-hole physics.

1. Introduction

Black holes represent the most extreme environments in the universe—regions where conventional descriptions of spacetime, matter, and information cease to function. In General Relativity (GR), a black hole is defined by the formation of an event horizon and the inevitable collapse of matter into a singularity of infinite density. Although this geometric picture has achieved remarkable observational success, it leaves several fundamental questions unresolved. Singularities violate physical law, the origin of black-hole entropy remains obscure, and the mechanism of Hawking radiation relies on semiclassical assumptions that do not provide a physical account of information flow. Moreover, gravitational lensing and shadow formation are treated as geometric consequences of curvature, without a deeper energetic or informational interpretation.

NEO-TEEM offers a different perspective. In this framework, the universe is described by three interchangeable energy states—vibrational energy v , condensed energy m , and spatial information capacity R . The ratio R/m , the information density, determines the local rate of vibrational evolution and governs gravitational behavior. Regions of low R/m suppress vibrational dynamics, slow the flow of time, and generate gravitational attraction. From this viewpoint, a black hole is not a geometric singularity but a

finite-information-density structure in which R/m approaches its minimum physically attainable value. This reinterpretation resolves several long-standing issues. Because R cannot collapse to zero, singularities do not form; instead, the interior becomes a saturated information-density core where vibrational evolution nearly halts. The event horizon emerges as a boundary layer where R/m changes extremely rapidly, producing the familiar “time-freezing” effect. Black-hole entropy arises from the constrained spatial information capacity of this boundary layer, naturally reproducing the area law. Hawking-like radiation appears as slow reconfiguration of R , not as pair creation in curved spacetime, ensuring that information is preserved. Gravitational lensing and shadow formation follow from spatial refraction: light bends wherever R decreases, unifying optical and gravitational phenomena under a single principle.

The purpose of this paper is to develop the full information-density model of black holes and lensing within the NEO-TEEM framework. We derive the structure of horizons, interior regions, and boundary layers; reinterpret entropy and evaporation in terms of R ; and show that lensing, photon spheres, and black-hole shadows arise from spatial variations in information capacity. This approach provides a physically transparent alternative to geometric curvature, reproducing the empirical successes of GR while offering new insights and testable predictions.

2. Information Density and the Collapse of R

In the NEO-TEEM framework, gravity and black-hole formation arise from the behavior of the information density

$$I = \frac{R}{m},$$

which measures the available spatial information capacity per unit of condensed energy. Unlike geometric curvature in General Relativity, information density is a physically interpretable quantity: it determines how freely vibrational modes can evolve, how fast time flows, and how strongly matter influences its surroundings. Regions with low R/m suppress vibrational evolution, slow the passage of time, and generate gravitational attraction. Black holes correspond to the extreme limit of this process, where R/m approaches its minimum attainable value.

2.1 Mass as a Local Suppressor of Spatial Information Capacity

Condensed energy m reduces the surrounding spatial information capacity R . This suppression is not geometric but energetic: localized energy confines information, decreasing the number of accessible microstates in its vicinity. The presence of mass therefore produces a spatial profile

$$R(r) = R_\infty - \Delta R(r),$$

where R_∞ is the asymptotic information capacity of empty space and $\Delta R(r)$ increases as one approaches the mass.

The gravitational potential emerges from the gradient of information density:

$$g(r) = -\nabla \left[\left(\frac{R}{m} \right) \right].$$

Thus, gravity is not a force but a flow toward regions of lower information density.

2.2 Collapse of R and the Onset of Strong Gravity

As mass becomes more concentrated, the suppression of R intensifies. When the local information capacity falls below a critical threshold, vibrational evolution slows dramatically. Because the local rate of time flow is proportional to v/R ,

$$t_{\text{local}} \propto \frac{v}{R},$$

a decreasing R directly produces gravitational time dilation.

In the limit where R approaches its minimum value R_{min} , vibrational evolution nearly halts:

$$\frac{v}{R} \rightarrow 0.$$

This is the physical origin of the “freezing of time” at the horizon. No geometric singularity is required; the phenomenon arises from the energetic structure of space.

2.3 Information-Density Collapse as the Formation of a Black Hole

A black hole forms when the information density

$$I(r) = \frac{R(r)}{m(r)}$$

falls to its minimum possible value I_{min} . At this point:

- R cannot decrease further (information capacity saturates)
- m cannot increase without bound (condensed energy cannot confine infinite information)
- the ratio R/m becomes constant throughout the interior

Thus, the interior of a black hole is not a singularity but a **finite-information-density core**. The horizon corresponds to the radius where the gradient $\nabla(R/m)$ becomes extremely steep, producing the strong gravitational field observed in GR.

This interpretation eliminates the need for infinite curvature and resolves the singularity problem.

2.4 Continuity of R and the Absence of Singularities

Because R represents the number of accessible spatial microstates, it cannot collapse to zero. A vanishing R would imply the absence of any physical degrees of freedom, contradicting the existence of a meaningful interior. Instead, R asymptotically approaches a finite minimum value determined by the energetic structure of the system.

This ensures:

- **no divergence of curvature**
- **no infinite density**
- **no breakdown of physical law**

The interior is a region of **saturated information density**, not a pointlike singularity.

2.5 Gravitational Strength as the Gradient of R/m

The gravitational field around a compact object is determined by how rapidly R/m changes with radius. A steep gradient produces strong lensing, time dilation, and the characteristic behavior of horizons. In the NEO-TEEM picture:

- Weak gravity corresponds to gentle variations in R/m
- Strong gravity corresponds to rapid collapse of R
- Black holes correspond to the limiting case where R/m reaches its minimum

This provides a unified physical mechanism for gravitational attraction, time dilation, and horizon formation.

2.6 Summary

The collapse of R is the fundamental process underlying black-hole formation. Mass suppresses spatial information capacity, decreasing R/m and slowing vibrational evolution. When R/m reaches its minimum value, a finite-information-density core forms, surrounded by a boundary layer where the gradient of R/m becomes extremely steep. This boundary layer is the event horizon. No singularities arise, and all gravitational phenomena—including lensing, time dilation, and shadow formation—follow from the behavior of information density.

3. Horizon as an Information-Density Boundary Layer

In General Relativity, the event horizon is defined geometrically as the surface beyond which light cannot escape. It is a null hypersurface, a purely mathematical boundary with no internal structure. In NEO-TEEM, the horizon acquires a concrete physical meaning: it is a **boundary layer in which the information density R/m undergoes its steepest spatial variation**. This boundary is not a geometric artifact but an energetic transition zone where the spatial information capacity collapses toward its minimum value.

The horizon forms because the local rate of vibrational evolution, proportional to v/R , approaches zero as R decreases. Time does not “flow” across this boundary; instead, it becomes progressively slower as the information capacity saturates. The horizon is therefore the surface where the suppression of R becomes strong enough that vibrational modes can no longer propagate outward. Light does not escape because the medium through which it would propagate has become too information-dense to support outward vibrational evolution.

3.1 The Horizon as the Steepest Gradient of R/m

The defining feature of a black hole in NEO-TEEM is the collapse of the information density

$$I(r) = \frac{R(r)}{m(r)}.$$

The horizon corresponds to the radius r_H where the radial derivative

$$|\nabla\left(\frac{R}{m}\right)|$$

reaches its maximum. This steep gradient produces:

- the strongest gravitational acceleration,
- the largest time dilation,
- the onset of vibrational freezing,
- and the effective trapping of light.

Thus, the horizon is not a point where “escape velocity equals c ” but a **physical transition layer** where the informational structure of space changes most rapidly.

3.2 Time Freezing as $v/R \rightarrow 0$

In NEO-TEEM, the local flow of time is governed by

$$t_{\text{local}} \propto \frac{v}{R}.$$

As R collapses toward its minimum value R_{min} , the ratio v/R decreases, slowing vibrational evolution. At the horizon:

$$\frac{v}{R} \rightarrow 0,$$

meaning that **time effectively halts** for any process attempting to propagate outward. This reproduces the GR prediction that clocks appear to stop at the horizon, but with a clear physical mechanism: **the medium becomes too information-dense to support vibrational change.**

3.3 Finite Thickness of the Horizon Layer

Unlike the infinitely thin geometric surface of GR, the NEO-TEEM horizon has a **finite thickness** determined by the spatial scale over which R/m transitions from its exterior profile to its interior saturated value. This thickness reflects the physical process of information-capacity collapse and has several consequences:

- the horizon is a **real physical region**, not a mathematical surface
- the boundary layer stores the majority of black-hole entropy
- Hawking-like radiation originates from fluctuations within this layer
- the photon sphere and shadow radius depend on the detailed shape of this layer

This finite-thickness structure is essential for resolving the singularity and information-loss problems.

3.4 Continuity of R and the Elimination of Singularities

Because R represents spatial information capacity, it cannot drop discontinuously or reach zero. The horizon marks the point where R approaches its minimum value smoothly. This continuity ensures:

- **no divergence of curvature,**
- **no infinite density,**

- no breakdown of physical law,
- no singularity.

The interior is a region of **saturated information density**, not an undefined geometric point.

3.5 Comparison with the GR Horizon

Feature	GR	NEO-TEEM
Definition	Null surface	Boundary layer of steepest R/m gradient
Thickness	Zero	Finite
Time behavior	Coordinate singularity	Physical freezing $v/R \rightarrow 0$
Entropy	Geometric area	Boundary information capacity
Interior	Singular	Finite-information core
Light trapping	Curvature	Collapse of R (spatial refraction)

NEO-TEEM preserves all observational predictions of GR—shadow size, lensing, time dilation—while providing a physically transparent mechanism for each.

3.6 Summary

The horizon is the **informational boundary** of a black hole: a finite-thickness layer where the information density R/m collapses most rapidly, suppressing vibrational evolution and trapping light. This reinterpretation replaces geometric singularities with energetic structure, resolves the paradoxes associated with horizons, and sets the stage for understanding entropy, evaporation, and lensing within a unified framework.

4. Interior Structure: Finite-Information Core

In General Relativity, the interior of a black hole collapses inevitably toward a singularity of infinite density and vanishing volume. This outcome reflects the geometric nature of GR: once curvature becomes sufficiently strong, nothing prevents the metric from diverging. In NEO-TEEM, the situation is fundamentally different. Because the spatial information capacity R cannot fall below a finite minimum value, and condensed energy m cannot confine infinite information, the ratio R/m saturates rather than diverges. The interior of a black hole is therefore not a singularity but a **finite-information-density core** in which vibrational evolution is nearly frozen and the informational structure of space reaches its most compressed state.

4.1 Saturation of R : The Minimum Information Capacity

The collapse of matter reduces the local spatial information capacity R , but this reduction cannot continue indefinitely. R represents the number of accessible microstates per unit region; it cannot reach zero without eliminating the physical degrees of freedom required for any meaningful interior. Instead, as collapse

proceeds:

$$R(r) \rightarrow R_{\min},$$

a finite, nonzero value determined by the energetic structure of the system. This saturation marks the transition from ordinary gravitational compression to the black-hole interior regime.

At this point:

- the information capacity cannot decrease further,
- the medium becomes maximally “stiff” in the informational sense,
- and vibrational modes cannot propagate.

This is the physical origin of the “frozen interior.”

4.2 Stabilization of m : No Divergence of Condensed Energy

Just as R cannot collapse to zero, condensed energy m cannot increase without bound. The confinement of information requires energy, and the energetic cost of compressing information grows rapidly as R approaches its minimum. This produces a natural stabilization:

$$m(r) \rightarrow m_{\text{core}},$$

a finite maximum density determined by the balance between information confinement and spatial capacity.

Thus:

- **no infinite density,**
- **no divergence of curvature,**
- **no singularity.**

The interior is a **stable, finite-density configuration**.

4.3 Constant Information Density in the Core

Once both R and m saturate, their ratio becomes constant:

$$\frac{R(r)}{m(r)} = I_{\min}.$$

This constant information density defines the interior of the black hole. Because $I_{\min}$ is the lowest physically attainable value, the interior cannot evolve further; vibrational modes are suppressed, and time flow is effectively halted:

$$\frac{v}{R_{\min}} \approx 0.$$

The interior is therefore a **static informational phase**, not a dynamic region of diverging curvature.

4.4 Physical Meaning of the Frozen Interior

The frozen interior has several important consequences:

- **Time evolution is nearly zero** $\rightarrow$ internal processes cannot proceed; the core is effectively timeless.
- **Vibrational modes cannot propagate** $\rightarrow$ no internal radiation, no oscillations, no turbulence.
- **Information is stored, not destroyed** $\rightarrow$ the interior acts as a reservoir of confined information.

- **The boundary layer (horizon) becomes the active region** → entropy, radiation, and lensing originate from the boundary, not the core.

This resolves the information-loss paradox: information is neither destroyed nor trapped in a singularity; it is stored in a finite-information core and gradually released through boundary-layer reconfiguration.

4.5 No Singularities: A Continuous R -Profile

Because both R and m vary smoothly and saturate at finite values, the interior profile is continuous:

$$R(r), m(r) \in C^1,$$

with no discontinuities or divergences. The collapse of information density is steep but finite, and the interior is a well-defined physical region.

This continuity ensures:

- no breakdown of physical law,
- no need for quantum gravity corrections,
- no infinite curvature,
- no geometric singularity.

The black hole is a **phase of matter-information compression**, not a geometric pathology.

4.6 Summary

The interior of a NEO-TEEM black hole is a **finite-information-density core** where both spatial information capacity R and condensed energy m saturate. The ratio R/m reaches its minimum value, vibrational evolution halts, and time effectively freezes. This structure replaces the singularity of GR with a physically meaningful, stable interior. The horizon becomes the active boundary layer where information is stored, entropy is defined, and radiation emerges. This reinterpretation resolves the singularity and information-loss problems while preserving all observable predictions of black-hole physics.

5. Black Hole Entropy and Boundary Information

Black-hole entropy is one of the most profound clues about the microscopic structure of spacetime. In General Relativity, the Bekenstein–Hawking formula

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A$$

assigns an entropy proportional to the horizon area, yet GR itself provides no explanation for *why* the degrees of freedom scale with area rather than volume, nor what physical quantity this entropy counts. In NEO-TEEM, the answer emerges naturally: **black-hole entropy is the information capacity of the boundary layer where the spatial information density R/m undergoes its steepest transition.**

The interior of a black hole is a finite-information-density core in which R and m saturate and R/m becomes constant. Because vibrational evolution is nearly frozen in this region, the interior contributes little to the accessible microstates. Instead, the horizon—the boundary layer where R collapses most

rapidly—contains the majority of the system’s informational degrees of freedom. The area law follows directly from the fact that this boundary layer is a two-dimensional surface whose thickness is set by the scale over which R transitions from its exterior profile to its interior saturated value.

5.1 Entropy as the Information Capacity of the Horizon Layer

In NEO-TEEM, entropy is identified with the growth of spatial information capacity R . For a black hole, the relevant quantity is the **excess information capacity stored in the boundary layer**, where the gradient of R/m is maximal. The number of accessible microstates is therefore proportional to the integral of R over this layer:

$$S \propto \int_{\text{horizon}} R dA.$$

Because the thickness of the layer is fixed by the saturation scale of R , the integral reduces to:

$$S \propto A,$$

reproducing the Bekenstein–Hawking area law without invoking geometric curvature or quantum gravity.

5.2 Why the Interior Does Not Contribute

The interior of a NEO-TEEM black hole is a region where:

- $R \rightarrow R_{\min}$,
- $m \rightarrow m_{\text{core}}$,
- $R/m = I_{\min}$ is constant,
- vibrational evolution $v/R \approx 0$.

Because the informational structure is saturated and static, the interior has **no accessible microstates**. All dynamical degrees of freedom reside in the boundary layer, where R is still capable of reconfiguration.

This explains:

- why entropy scales with area, not volume
- why the interior does not store thermodynamic information
- why information is not destroyed inside the black hole

The interior is a *frozen phase*, not a reservoir of entropy.

5.3 Boundary Degrees of Freedom and the Area Law

The area law emerges because the boundary layer is effectively two-dimensional. Its thickness is determined by the scale over which R transitions from its exterior value to $R_{\min}$. This thickness is universal and does not scale with black-hole mass. Therefore:

- the number of degrees of freedom is proportional to the surface area
- the entropy is proportional to the area
- the coefficient is set by the microscopic scale of R -variation

This provides a physical explanation for the universality of the Bekenstein–Hawking formula.

5.4 Entropy as Constrained Spatial Information

In NEO-TEEM, entropy is not an abstract statistical quantity but a measure of **how much spatial information is constrained**. At the horizon:

- the collapse of R is steep
- the information capacity is highly compressed
- the boundary layer stores the “missing” information that cannot be accommodated in the interior

Thus, black-hole entropy counts the number of ways the boundary layer can reconfigure while maintaining the same macroscopic parameters (mass, charge, spin).

This interpretation is consistent with:

- holographic principles
- membrane-paradigm intuition
- the thermodynamic behavior of horizons

but arises from a concrete energetic mechanism rather than geometric assumptions.

5.5 Connection to Hawking-Like Radiation

Because the boundary layer contains the accessible microstates, **Hawking-like radiation originates from fluctuations in this layer**, not from pair creation in curved spacetime. Small reconfigurations of R allow vibrational energy to leak outward, producing a nearly thermal spectrum. Information is preserved because the boundary layer evolves continuously:

$$\Delta S \propto \Delta R_{\text{boundary}}.$$

Thus:

- entropy decrease corresponds to boundary-layer relaxation
- radiation carries information encoded in R
- evaporation is a slow release of constrained spatial information

This resolves the information-loss paradox without modifying quantum mechanics.

5.6 Summary

Black-hole entropy in NEO-TEEM arises from the **information capacity of the horizon boundary layer**, where the spatial information density R/m collapses most rapidly. The interior contributes no entropy because it is a saturated, frozen informational phase. The area law follows naturally from the two-dimensional nature of the boundary layer, and Hawking-like radiation emerges from its slow reconfiguration. This provides a physically transparent, non-geometric explanation for black-hole thermodynamics and unifies entropy, horizons, and evaporation under a single informational principle.

6. Hawking-Like Radiation as R -Reconfiguration

In General Relativity, Hawking radiation is derived from quantum field theory on a curved background, where particle–antiparticle pairs form near the horizon and one escapes to infinity. While this semiclassical

picture reproduces the thermal spectrum, it does not explain how information is preserved, nor does it identify a physical mechanism for the energy transfer. In NEO-TEEM, Hawking-like radiation arises from a different and more fundamental process: **slow reconfiguration of the spatial information capacity R in the horizon boundary layer.**

Because the interior of a black hole is a saturated information-density core where $R/m = I_{\min}$ and vibrational evolution is nearly frozen, the only region capable of dynamical change is the boundary layer where R transitions from its exterior value to $R_{\min}$. Small fluctuations in this layer allow vibrational energy to leak outward. Radiation is therefore not created from vacuum fluctuations but emerges from **energetic relaxation of the boundary information structure.**

This mechanism produces a spectrum that is approximately thermal at large scales but retains subtle correlations that encode the internal information of the black hole. Evaporation is thus a slow, continuous release of constrained spatial information, not a loss of quantum coherence.

6.1 Boundary-Layer Fluctuations as the Source of Radiation

The horizon boundary layer contains the majority of the black hole's accessible microstates. Because this layer has finite thickness and finite information capacity, it can undergo small reconfigurations:

$$R_{\text{boundary}} \rightarrow R_{\text{boundary}} + \delta R.$$

These fluctuations modify the local ratio v/R , allowing a small amount of vibrational energy to escape. The escaping radiation is therefore:

- **not pair creation,**
- **not vacuum polarization,**
- **but the outward leakage of vibrational modes enabled by changes in R .**

This provides a physically transparent mechanism for Hawking-like emission.

6.2 Nearly Thermal Spectrum from Statistical R -Fluctuations

Because the boundary layer contains a large number of microstates, its fluctuations are statistically distributed. The resulting radiation spectrum is:

- **approximately thermal,**
- **but not exactly Planckian,**
- **with small deviations encoding information.**

The temperature is determined by the gradient of R/m at the horizon:

$$T \propto \left| \nabla \left(\frac{R}{m} \right) \right|_{r_H}.$$

This reproduces the Hawking temperature formula in the appropriate limit while providing a physical interpretation of its origin.

6.3 Information Preservation Through Continuous R -Evolution

Because radiation arises from reconfiguration of R , not from particle creation, information is never lost.

The boundary layer evolves continuously:

$$\Delta S \propto \Delta R_{\text{boundary}},$$

and the emitted radiation carries correlations that reflect the microscopic state of the horizon. This resolves the information-loss paradox without modifying quantum mechanics or invoking nonlocality.

Key points:

- **Information is stored in the boundary layer**
- **Radiation is produced by boundary-layer relaxation**
- **The emitted quanta inherit correlations from R**
- **No singularity exists to destroy information**

Thus, evaporation is a **unitary process**.

6.4 Evaporation as Slow Relaxation of the Horizon Structure

As radiation escapes, the boundary layer gradually loses information capacity:

$$R_{\text{boundary}}(t) \downarrow.$$

This causes:

- a decrease in black-hole mass m ,
- a reduction in the steepness of the R/m gradient,
- a corresponding increase in the effective temperature,
- and an acceleration of the evaporation process.

The final stages of evaporation correspond to the collapse of the boundary layer itself, not to a catastrophic quantum-gravity event. Because the interior is a finite-information core, the black hole simply transitions out of the black-hole phase as R/m rises above the critical threshold.

6.5 No Firewalls, No Singularities, No Paradoxes

The NEO-TEEM mechanism avoids all major conceptual problems associated with Hawking radiation:

- **No firewall** → the boundary layer is a physical region with finite thickness and finite temperature.
- **No singularity** → the interior is a finite-information core.
- **No information loss** → radiation encodes boundary-layer correlations.
- **No need for quantum gravity** → the mechanism is energetic and informational, not geometric.

This provides a unified, physically grounded picture of black-hole evaporation.

6.6 Summary

Hawking-like radiation in NEO-TEEM arises from **reconfiguration of the spatial information capacity** R in the horizon boundary layer. Small fluctuations in this layer allow vibrational energy to escape, producing a nearly thermal spectrum that carries information. Evaporation is a slow relaxation of the boundary structure, not a quantum-field-theoretic pair-creation process. This mechanism preserves unitarity, eliminates singularities, and provides a coherent physical explanation for black-hole thermodynamics.

7. Spatial Refraction and Gravitational Lensing

In General Relativity, gravitational lensing is interpreted as the bending of light caused by the curvature of spacetime. In NEO-TEEM, the same phenomenon arises from a different and more physically transparent mechanism: **spatial refraction produced by gradients in the information density R/m** . Light propagates as a vibrational excitation whose speed and trajectory depend on the local spatial information capacity. When R decreases near a massive object, the medium becomes more “information-dense,” slowing vibrational evolution and bending the path of light toward regions of lower R/m .

This mechanism unifies optical refraction in materials and gravitational lensing in vacuum. Both arise from the same principle: **light bends wherever the spatial information capacity changes**.

7.1 Refractive Index of Space: $n(r) = R_{vac}/R(r)$

In NEO-TEEM, the effective refractive index of space is determined by the ratio of the information capacity of vacuum to that of the local region:

$$n(r) = \frac{R_{vac}}{R(r)}.$$

As mass suppresses R , the refractive index increases:

- **far from mass:** $R \approx R_{vac}$, so $n \approx 1$
- **near mass:** $R < R_{vac}$, so $n > 1$
- **near horizon:** $R \rightarrow R_{min}$, so $n \rightarrow n_{max}$

Thus, the gravitational field acts as a **gradient-index medium** (GRIN medium), exactly like optical materials whose refractive index varies in space.

This provides a direct physical mechanism for gravitational lensing.

7.2 Light Trajectories as Extremal Paths of v/R

Light follows the path that maximizes the local vibrational evolution rate. Because the speed of vibrational propagation is proportional to v/R , the trajectory satisfies:

$$\delta \int n(r) ds = 0,$$

which yields a generalized Snell’s law:

$$n(r) \sin \theta = \text{constant}.$$

This applies equally to:

- optical interfaces
- graded-index materials
- gravitational fields
- black-hole environments

Thus, **gravitational lensing is a form of spatial refraction**.

7.3 Comparison with GR: Same Predictions, Different Mechanism

Although the mechanism differs, NEO-TEEM reproduces all observational predictions of GR:

- deflection angle around stars
- Einstein rings
- weak-lensing shear fields
- strong lensing by clusters
- photon-sphere radius
- black-hole shadow size

The equivalence arises because the GR curvature profile and the NEO-TEEM refractive-index profile produce the same extremal paths.

But NEO-TEEM provides a **physical** explanation:

- curvature $\rightarrow$ geometric description
- refraction $\rightarrow$ energetic/informational description

NEO-TEEM explains *why* light bends.

7.4 Photon Sphere as an Information-Density Trap

The photon sphere corresponds to the radius where the refractive index gradient is strong enough to trap light in circular orbits:

$$\frac{d}{dr}(n(r) r) = 0.$$

Because $n(r)$ diverges as $R \rightarrow R_{\min}$, the photon sphere arises naturally as the point where:

- the inward pull from decreasing R
- balances
- the outward geometric divergence of the path

This reproduces the GR photon-sphere radius without invoking curvature.

7.5 Shapiro Delay as Propagation Through Low- R Regions

The Shapiro time delay arises because light slows in regions where R is suppressed:

$$t_{\text{travel}} \propto \int \frac{n(r)}{c} ds.$$

As $n(r)$ increases near mass, the travel time increases. This is identical to optical delay in high-index media.

Thus:

- **Shapiro delay = optical delay in a gravitational GRIN medium**
- **No curvature is required**
- **The effect follows directly from the collapse of R**

7.6 Lensing Near Horizons: Extreme Spatial Refraction

Near the horizon, R approaches its minimum value and the refractive index becomes extremely large:

$$n(r) \rightarrow n_{\max}.$$

This produces:

- extreme bending of light
- strong asymmetry in lensing patterns
- the characteristic brightness crescent in EHT images
- the sharp boundary of the black-hole shadow

The shadow is therefore the **optical consequence of the R-collapse**, not a geometric silhouette.

7.7 Summary

Gravitational lensing in NEO-TEEM is a manifestation of **spatial refraction** caused by gradients in the information density R/m . Mass suppresses R , increasing the refractive index and bending light toward regions of lower information capacity. This mechanism reproduces all predictions of GR while providing a unified explanation for optical refraction, gravitational lensing, photon spheres, Shapiro delay, and black-hole shadows. Light does not follow curved spacetime—it follows the informational structure of space.

8. Black Hole Shadows and Observational Signatures

Black-hole shadows provide a direct observational probe of the informational structure of spacetime. In General Relativity, the shadow is interpreted as the projection of unstable photon orbits shaped by spacetime curvature. In NEO-TEEM, the shadow arises from **spatial refraction in regions where the information density R/m collapses toward its minimum value**. The refractive index $n(r) = R_{\text{vac}}/R(r)$ increases sharply near the horizon, bending light inward and producing a dark region surrounded by a bright photon ring.

This interpretation preserves all empirical successes of GR while offering a physically transparent mechanism for shadow formation. Moreover, the NEO-TEEM model predicts subtle deviations from GR that may be detectable with next-generation interferometers.

8.1 Shadow Radius from the R -Profile

The radius of the shadow is determined by the extremal behavior of the refractive index:

$$n(r) = \frac{R_{\text{vac}}}{R(r)}.$$

The shadow boundary corresponds to the radius where the deflection angle diverges, which occurs near the maximum of:

$$n(r) r.$$

Because $R(r)$ decreases more steeply near the horizon than in GR's curvature profile, NEO-TEEM predicts:

- a **slightly larger shadow radius** for the same mass,
- a **smoother transition** between the photon ring and the shadow interior,
- and a **marginally thicker photon ring**.

These differences are small but potentially measurable.

8.2 Photon Ring Structure from Information-Density Gradients

The photon ring arises from light that orbits multiple times near the radius where:

$$\frac{d}{dr}(n(r) r) = 0.$$

In NEO-TEEM:

- the steepness of the R -collapse determines the sharpness of the ring,
- the finite thickness of the horizon layer broadens the ring slightly,
- and the nonlinearity of $R(r)$ introduces **azimuthal asymmetries**.

These asymmetries match features observed in EHT images of M87* and Sgr A*, where the brightness crescent is not perfectly symmetric.

8.3 Asymmetry and Brightness Crescent

Because the refractive index depends on the local information density, any anisotropy in R caused by:

- accretion flow,
- magnetic fields,
- or asymmetric infall history

produces **direction-dependent bending**.

This naturally yields:

- a **bright crescent** on the approaching side,
- a **dimmer region** on the receding side,
- and **noncircular distortions** in the shadow boundary.

NEO-TEEM therefore explains the EHT brightness asymmetry without requiring extreme Doppler boosting.

8.4 Lensing Profiles and Deviations from GR

Because gravitational lensing is spatial refraction, the lensing profile depends on the detailed shape of $R(r)$.

NEO-TEEM predicts:

- slightly stronger bending at intermediate radii,
- slightly weaker bending extremely close to the horizon (due to finite-thickness effects),
- and small deviations in the magnification pattern of strong-lensing arcs.

These deviations are within the sensitivity of future VLBI arrays and space-based interferometers.

8.5 Time-Delay Signatures (*Shapiro and Beyond*)

The Shapiro delay arises from propagation through regions of high refractive index:

$$t_{\text{travel}} \propto \int n(r) ds.$$

NEO-TEEM predicts:

- the same qualitative delay as GR,

- but with **slightly enhanced delays** near the horizon due to the sharper collapse of R ,
- and **frequency-dependent corrections** because R affects vibrational modes differently.

These effects could be probed by pulsar timing near Sgr A*.

8.6 Multi-Messenger Signatures

Because the boundary layer has finite thickness and stores information, NEO-TEEM predicts:

- **long-lived echoes** in gravitational-wave ringdown,
- **correlated fluctuations** in near-horizon emission,
- **nonthermal features** in late-stage evaporation,
- and **polarization signatures** from anisotropic R -gradients.

These provide multiple observational pathways to test the theory.

8.7 Summary

Black-hole shadows and lensing phenomena in NEO-TEEM arise from **spatial refraction in regions where the information density R/m collapses**. This mechanism reproduces all observational successes of GR while predicting subtle deviations in shadow radius, photon-ring structure, brightness asymmetry, lensing profiles, and time-delay signatures. These effects offer concrete, testable predictions for current and future observational facilities.

9. Connection to Cosmology: Local vs Global R

Black holes and cosmic expansion appear to represent opposite extremes of gravitational behavior: one collapses spacetime, the other drives it apart. In NEO-TEEM, these phenomena arise from the same underlying mechanism—the evolution of spatial information capacity R . A black hole forms when R collapses locally toward its minimum value, while cosmic expansion occurs because R increases globally as vibrational energy converts into spatial information. The universe is therefore governed by a single informational principle whose local and global manifestations produce the full range of gravitational phenomena.

This unified perspective reveals that black holes are not exceptions to cosmology but integral components of the same informational dynamics. The tension between local and global behavior of R explains observational puzzles such as the H_0 discrepancy and provides a coherent framework for understanding structure formation, lensing, and the cosmic expansion history.

9.1 Local Collapse of R vs Global Growth of R

The key distinction between black holes and cosmic expansion is the **sign** of the evolution of R :

- **Black holes:**

$$\frac{dR}{dt} < 0 (\text{local collapse})$$

Mass suppresses R , driving it toward $R_{\min}$.

- **Cosmic expansion:**

$$\frac{dR}{dt} > 0 (\text{global growth})$$

Vibrational energy converts into spatial information, increasing R and producing expansion.

Thus, black holes and cosmology are **two directions of the same informational flow**.

9.2 Black Holes as Local “Negative Expansion”

In cosmology, the expansion rate is governed by the global behavior of R :

$$H \propto \frac{1}{R} \frac{dR}{dt}.$$

Inside a black hole, the same expression applies but with opposite sign:

$$H_{\text{local}} \propto \frac{1}{R} \frac{dR}{dt} < 0.$$

This means:

- black holes are **regions of negative expansion**,
- the interior is a **contracting informational phase**,
- the horizon marks the transition between global expansion and local collapse.

This duality is impossible in GR but natural in NEO-TEEM.

9.3 The Cosmic Tension Landscape

Because R varies across space, the universe contains a **tension landscape**:

- regions of high R (voids) $\rightarrow$ faster local expansion
- regions of low R (clusters) $\rightarrow$ slower expansion
- extreme low- R regions (BHs) $\rightarrow$ collapse

This landscape explains:

- **H_0 tension** $\rightarrow$ local measurements probe low- R environments
- **BAO discrepancies** $\rightarrow$ intermediate-scale R -gradients
- **SN Ia consistency** $\rightarrow$ probes large-scale average $R(t)$

Black holes are simply the deepest wells in this landscape.

9.4 Information Flow Between Local and Global R

Because R is a conserved informational quantity at the global level, local collapse must be balanced by global growth. This leads to a natural interpretation:

- black holes **store** spatial information in compressed form
- cosmic expansion **releases** spatial information globally
- the universe evolves by redistributing information between these phases

This provides a thermodynamic interpretation of cosmic evolution.

9.5 Black Holes as Seeds of Structure in an R -Driven Universe

In NEO-TEEM, black holes play an essential cosmological role:

- they anchor the deepest R -gradients
- they shape lensing fields
- they regulate local expansion rates
- they influence the global $R(t)$ evolution through information confinement

Thus, black holes are not isolated objects but **active participants in the cosmic information cycle**.

9.6 Observational Consequences

The connection between local and global R predicts:

- **environment-dependent H_0 values**
- **anisotropic lensing signatures**
- **correlations between BH demographics and cosmic expansion history**
- **R -dependent deviations in BAO and CMB lensing**
- **subtle shifts in shadow size depending on cosmological $R(t)$**

These effects provide new observational tests of NEO-TEEM.

9.7 Summary

Black holes and cosmic expansion are two manifestations of the same informational dynamics. Black holes represent **local collapse of R** , while cosmology reflects **global growth of R** . The tension landscape created by spatial variations in R explains observational discrepancies and unifies gravitational phenomena across all scales. In NEO-TEEM, the universe is a single informational system whose local and global behaviors are deeply interconnected.

10. Predictions and Tests

A physical theory must make predictions that can be tested. NEO-TEEM provides a unified informational description of gravity, black holes, and cosmic expansion, and this framework yields a set of concrete, falsifiable predictions. Some predictions reproduce the successes of General Relativity, while others introduce subtle deviations that can be probed with current or near-future observational facilities. These signatures arise from the behavior of the spatial information capacity R and its gradients, which govern lensing, time dilation, horizon structure, and radiation.

The predictions fall into four categories:

1. **Shadow and lensing signatures**
2. **Time-delay and photon-sphere deviations**
3. **Gravitational-wave and horizon-echo phenomena**
4. **Cosmological correlations with local R -gradients**

Together, these provide a comprehensive observational program for testing the NEO-TEEM framework.

10.1 Shadow Radius Deviations

NEO-TEEM predicts that the shadow radius is determined by the extremal behavior of the refractive index:

$$n(r) = \frac{R_{\text{vac}}}{R(r)}.$$

Because $R(r)$ collapses more steeply near the horizon than the GR curvature profile, the theory predicts:

- a **slightly larger shadow radius** (1–3% for typical supermassive BHs)
- a **smoother transition** between the photon ring and shadow interior
- a **marginally thicker photon ring**

These deviations are small but measurable with next-generation EHT baselines and space-VLBI.

10.2 Photon-Ring Asymmetry and Noncircularity

The nonlinearity of the R -profile near the horizon produces:

- **azimuthal asymmetry** in the photon ring
- **noncircular distortions** in the shadow boundary
- **brightness crescents** that do not require extreme Doppler boosting

These features match current EHT observations qualitatively but predict **specific patterns** in higher-resolution images.

10.3 Frequency-Dependent Lensing

Because the effective refractive index depends on the vibrational mode, NEO-TEEM predicts:

- **weak frequency dependence** in strong lensing
- **chromatic distortions** near the photon sphere
- **slightly different shadow radii** for radio vs sub-mm wavelengths

This is a unique signature not present in GR.

10.4 Enhanced Shapiro Delay Near Horizons

The collapse of R near the horizon increases the refractive index, producing:

- **slightly larger Shapiro delays**
- **nonlinear corrections** at small impact parameters
- **frequency-dependent time delays**

Pulsar timing near Sgr A* could detect these effects.

10.5 Horizon Echoes in Gravitational Waves

Because the horizon has finite thickness, not a sharp geometric boundary, NEO-TEEM predicts:

- **late-time echoes** in the ringdown phase
- **frequency-dependent damping**

- **small phase shifts** relative to GR templates

These signatures are within reach of LIGO/Virgo/KAGRA and especially LISA.

10.6 Evaporation Deviations from Pure Thermality

Hawking-like radiation arises from boundary-layer reconfiguration, not pair creation. Therefore:

- the spectrum is **nearly thermal**,
- but contains **correlations** encoding boundary-layer microstructure,
- and exhibits **nonthermal tails** at late times.

These deviations preserve unitarity and provide a potential observational signature for primordial black holes.

10.7 Correlations Between Black-Hole Environments and H_0 Measurements

Because R varies across cosmic environments:

- regions with deep R -wells (clusters, BH-rich environments) show **slower local expansion**,
- voids show **faster local expansion**,
- leading to **environment-dependent H_0 values**.

This provides a natural explanation for the H_0 tension and predicts:

- **correlations between lensing fields and local H_0**
- **anisotropies in expansion rate aligned with R -gradients**

These correlations are testable with SN Ia, BAO, and lensing surveys.

10.8 Distinctive Lensing Profiles Around Intermediate-Mass Black Holes

NEO-TEEM predicts:

- **stronger bending** at intermediate radii
- **weaker bending** extremely close to the horizon
- **non-GR magnification patterns** in strong-lensing arcs

These effects are ideal for IMBH searches.

10.9 Summary of Testable Predictions

NEO-TEEM makes the following falsifiable predictions:

- Slightly enlarged shadow radius
- Photon-ring asymmetry from R -nonlinearity
- Frequency-dependent lensing
- Enhanced Shapiro delay
- Horizon echoes in gravitational waves
- Nonthermal correlations in evaporation
- Environment-dependent H_0 values
- Distinctive IMBH lensing signatures

These predictions provide a clear observational pathway for confirming or refuting the theory.

11. Conclusion

Black holes in the NEO-TEEM framework are not geometric singularities but **finite-information-density structures** formed by the collapse of spatial information capacity R . The horizon emerges as a **boundary layer** where the ratio R/m undergoes its steepest transition, suppressing vibrational evolution and producing the familiar freezing of time. The interior is a **saturated informational phase**, not a pointlike singularity, and the boundary layer stores the accessible microstates that give rise to black-hole entropy.

This informational interpretation reproduces all observational successes of General Relativity—lensing, time dilation, photon spheres, and shadow formation—while providing a physically transparent mechanism for each. Gravitational lensing becomes spatial refraction in a gradient-index medium defined by $n(r) = R_{\text{vac}}/R(r)$. Hawking-like radiation arises from slow reconfiguration of the boundary-layer information structure, ensuring that information is preserved and eliminating the paradoxes associated with semiclassical pair-creation models.

The NEO-TEEM description of black holes connects naturally to cosmology. Black holes represent **local collapse of R** , while cosmic expansion reflects **global growth of R** . These processes are two directions of the same informational dynamics, forming a unified picture of gravitational behavior across all scales. The resulting tension landscape explains environment-dependent expansion rates and provides a coherent framework for interpreting the H_0 discrepancy.

The theory yields a set of concrete, falsifiable predictions: slight deviations in shadow radius, photon-ring asymmetry, frequency-dependent lensing, enhanced Shapiro delay, horizon echoes in gravitational waves, and correlations between local R -gradients and cosmological observables. These signatures offer multiple observational pathways for confirming or refuting the NEO-TEEM model.

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